Stock Price Simulation using Geometric Brownian Motion

A Monte Carlo Method Implementation in C with Interactive Web Dashboard

Overview & Mathematical Model

Geometric Brownian Motion (GBM) models stock prices using the stochastic differential equation:

$$dS = \mu S\, dt + \sigma S\, dW$$

where μ is the expected return, σ is the volatility, and W represents a standard Brownian motion.
The C implementation utilizes the exact analytical solution derived from Itô's lemma, ensuring that simulated price paths carry no time-discretization error.

Implementation Details

Random numbers are generated using the **xoshiro256**** generator and transformed into normally distributed variables via the **Box-Muller transform**.
For every simulation request, the C engine processes thousands of price paths to compute:

- Percentile bands and terminal price distributions
- Value-at-Risk (VaR) and Expected Shortfall (ES)
- European option pricing using antithetic variates, validated against the Black-Scholes formula and put-call parity
- Convergence verification confirming Monte Carlo error reduction proportional to $1/\sqrt{N}$

Interactive Dashboard & Architecture

The C core doubles as a lightweight local web server, serving parameters and returning Monte Carlo results as JSON payloads to the browser dashboard.
Through the interactive dashboard, users can adjust key simulation variables in real time—including drift, volatility, time horizon, number of paths, and option strikes—and observe dynamic, animated chart visualizations.

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